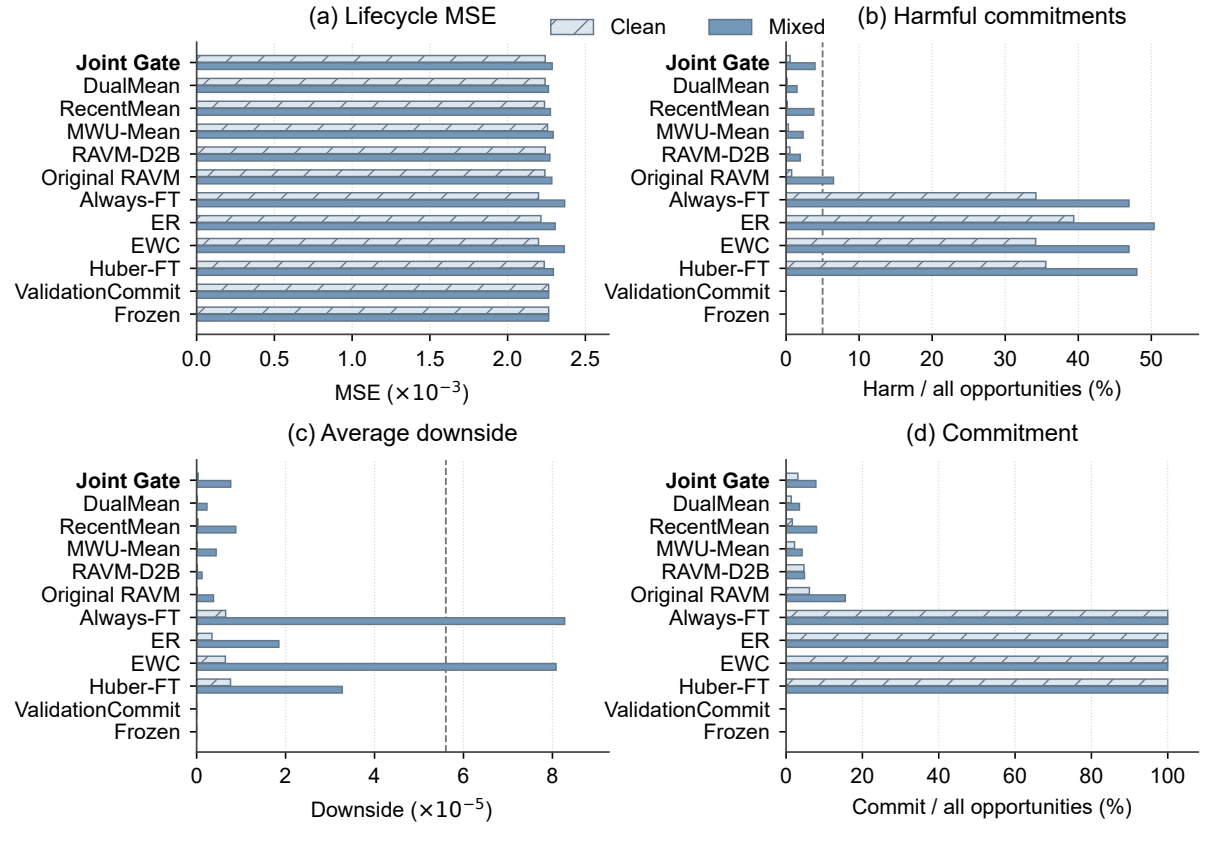
(a) Lifecycle MSE
Clean
Mixed
(b) Harmful commitments
Joint Gate
DualMean
RecentMean
MWU-Mean
RAVM-D2B
Original RAVM
Always-FT
ER
EWC
Huber-FT
ValidationCommit
Frozen
0.0
0.5
1.0
1.5
2.0
2.5
MSE (×10⁻³)
0
10
20
30
40
50
Harm / all opportunities (%)
(c) Average downside
(d) Commitment
0
2
4
6
8
Downside (×10⁻⁵)
0
20
40
60
80
100
Commit / all opportunities (%)